

C4 Exploratory Results Report

September 21, 2026 | Group discussion draft | Aligned with C4 integrated protocol v3

1. Executive findings and completed scope

Stages 0 and 1, and the two prespecified Stage 2 update forks, are complete. Proxy optimization frequently disagrees with labelled local evaluation. Both prespecified long-horizon endpoints exhibit an adverse feedback interaction on fixed-reference cross-entropy (CE). Persistent self-deterioration, a stability boundary, and a phase transition have not been established.

The candidate after block 5 immediately reduced CE on the fixed reference panel Q by 0.04593. After 1,410 subsequent updates, accepting it under endogenous supervision instead produced CE 0.26629 higher than rejecting it. Under a common supervision tape generated by the reject branch, acceptance remained beneficial by 0.05069. However, intermediate effects repeatedly changed sign: the endpoint does not describe the entire trajectory.

C4 stage	Protocol requirement	Actual status
Stage 0	Algorithm, data isolation, logs, recovery, timing	Complete; 12 software checks and real-data preflight passed
Stage 1	Two feedback endpoints, one seed, two cycles	Complete; 30 blocks and 2,820 updates per arm
Stage 2	Prespecified accept/reject and closed/replay forks	Complete; after blocks 5/15, H=20/94/1410
Stage 3	Perturbations, finite responses, closure and Jacobian checks	Not run; existing trajectories do not replace this stage
Stage 4	Frozen analysis, new seeds, untouched confirmation audit	Not run; confirmation outcomes remain uninspected

This is C4-natural / Type B: student → EMA teacher → pseudo-labels and confidence selection → next student. Images are exogenous; supervision and selection masks are endogenous. There is no label-based acceptance gate. Executing an update does not mean an evaluator certified its validity. There is no mixture-to-single-heir compression step; inheritance loss and Bayesian hypercompression are not applicable.

Frozen experimental settings

- CIFAR-100-C, severity 5, all 15 corruptions in a fixed order, repeated twice; seed 101.
- Source: Hendrycks2020AugMix_ResNeXt, 6,900,132 parameters. All student weights train; BN running statistics stay fixed. No stochastic source-weight restoration. This is not full CoTTA.
- Feedback vector $\gamma=(\lambda,\beta)$: only (0,0.001) and (1,0.001). The code's scalar gamma denotes λ . At $\lambda=0$, training continues with fixed-source supervision; $\lambda=1$ uses the evolving EMA teacher.
- SGD: learning rate 0.001, momentum 0.9, weight decay 0; batch size 64; confidence threshold 0.9; one update per nonempty batch.
- Stratified base-image ID split: adaptation 6,000, reserved gate 1,000, exploratory audit 1,500, confirmation audit 1,500. Labels never decide natural adaptation or acceptance.
- Q: the smallest exploratory-audit ID per class, crossed with all 15 corruptions: 100 base images and 1,500 fixed image-corruption pairs. This is an exploratory panel, not an independent confirmation test.

2. Stage 0 and C4-1: implementation and local validity

Both arms completed all 2,820 scheduled updates, with no empty-selection batches. A clean-source sanity check on 1,000 adaptation-pool IDs gave 20.8% error and CE 0.78928. This checks the loading path; it is not a new benchmark result.

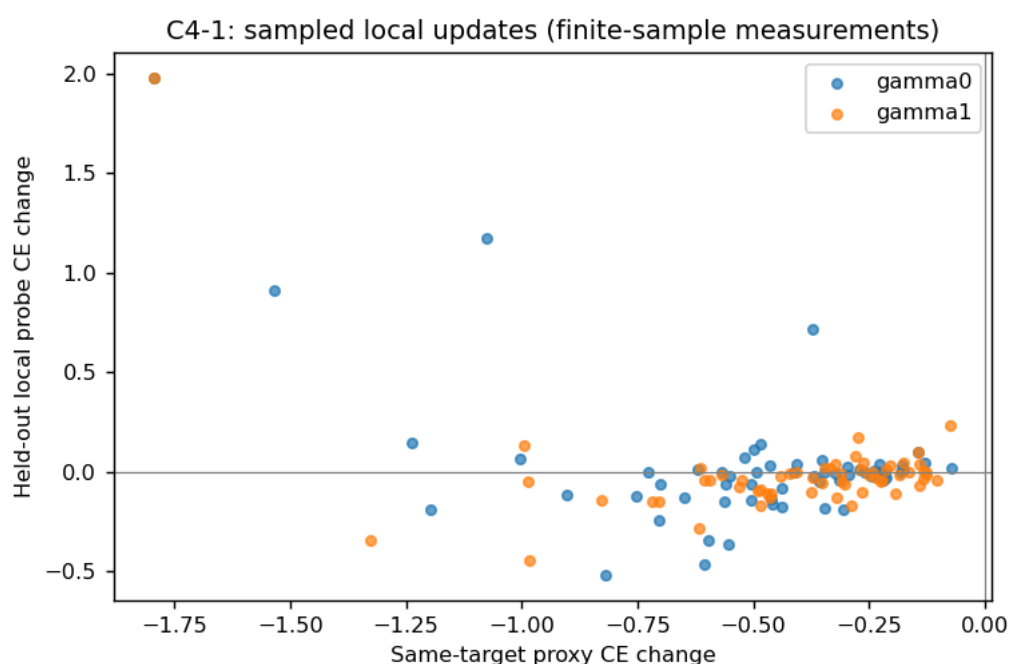
A uint8 label-indexing bug was repaired before real-data adaptation; shared optimizer momentum across fork restores was repaired before scientific fork runs. Failed preflight artifacts remain separate. Data integrity, full-state restoration, fixed-target loss recomputation, retained records, and replay consistency checks passed. Reported scientific results use the repaired implementation.

Does reducing the proxy improve the model locally?

Proxy changes use identical images, strong views, pseudo-labels, masks and normalization before/after the update, before EMA refresh. Labelled probes run on the first and last scheduled minibatch of each block, using the same 128 exploratory-audit IDs: 60 probes per arm. The sign tolerance is 10^{-6} .

Measurement	$\lambda=0$	$\lambda=1$
All updates: proxy CE decreases	2,815 / 2,820	2,817 / 2,820
All updates: proxy CE increases	5 / 2,820	3 / 2,820
Probes: proxy down, local audit CE down	38 / 60	43 / 60
Probes: proxy down, local audit CE up	22 / 60	17 / 60
Probes: proxy up, local down / local up	0 / 0	0 / 0
Probes: near-zero / empty mask	0 / 0	0 / 0

Observation: 36.7% of scheduled probes at $\lambda=0$ and 28.3% at $\lambda=1$ improve the proxy while worsening labelled local CE. This supports a finite-sample evaluation-performance gap. These rates cannot be extrapolated to all updates, and no population-validity guarantee follows. The gap occurs under both supervision sources, so it alone does not identify endogenous feedback as its cause.



C4-1. Proxy change versus labelled local CE change at all scheduled probes.

3. Stage 1 / C4-2b: performance on fixed Q

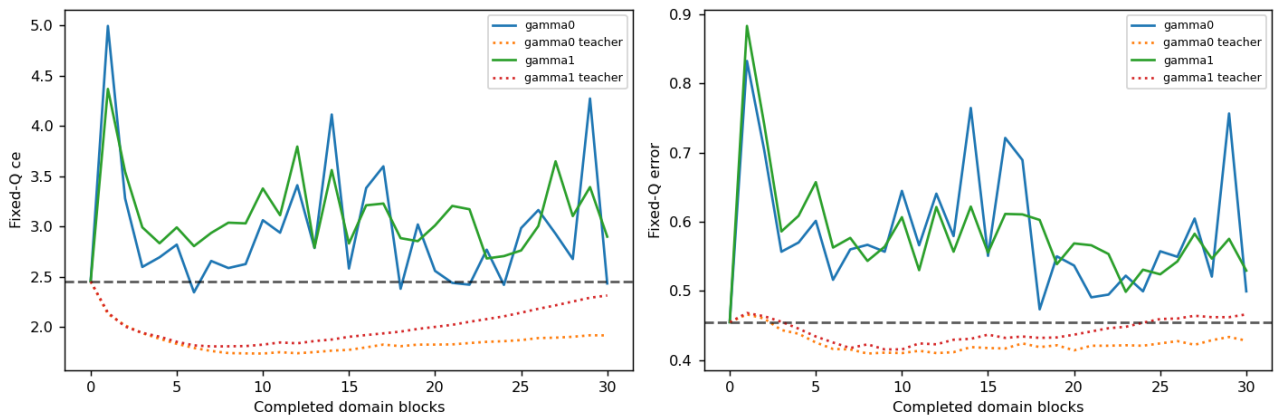
Every entry below uses the same Q. CE is the protocol's primary mechanism/validity readout; classification error is its benchmark endpoint. Both are retained. The frozen source remains at CE 2.45255 and error 45.47% throughout.

Time / model	$\lambda=0$ CE	$\lambda=0$ error	$\lambda=1$ CE	$\lambda=1$ error
Initial student / teacher	2.45255	45.47%	2.45255	45.47%
End of cycle 1: student	2.58033	55.07%	2.83018	55.53%
End of cycle 2: student	2.43198	49.93%	2.89772	52.93%
End of cycle 1: teacher	1.77089	41.73%	1.90086	43.67%
End of cycle 2: teacher	1.91454	42.80%	2.31199	46.60%

Student: At $\lambda=1$, endpoint CE exceeds the source and rises between cycle endpoints, while error falls. At $\lambda=0$, CE recovers after cycle 1 and ends slightly below the source, but error remains higher. Full trajectories fluctuate and recover; they do not establish persistent deterioration.

Teacher: EMA teacher CE is lower than student CE at both cycle endpoints in both arms. At $\lambda=1$, teacher CE rises from 1.90086 to 2.31199 but remains below source CE; teacher error ends at 46.60%. Conclusions depend on the evaluated model and loss. Student deterioration cannot be substituted for a claim that the whole system deteriorates. The teacher-student gap is not an inheritance loss here.

C4-2b: fixed reference distribution, same 1,500 image-corruption pairs



C4-2b. Student, teacher and source on the same fixed panel. Plot labels gamma0/1 denote $\lambda=0/1$ with β fixed.

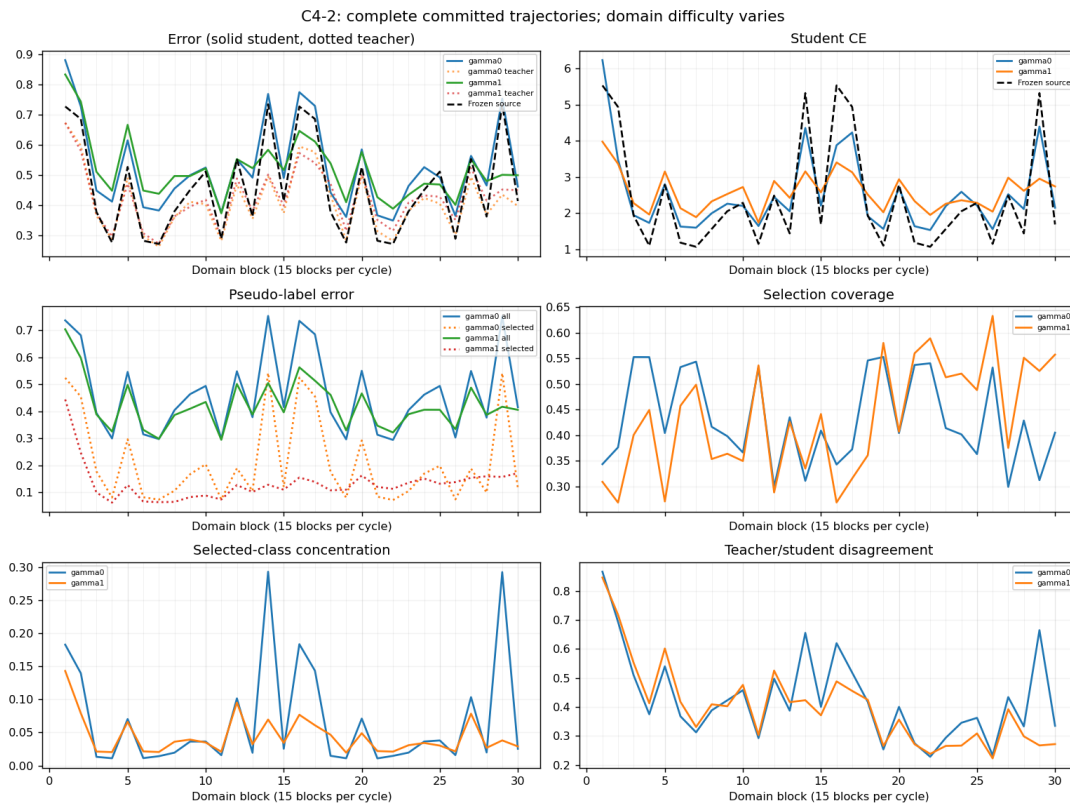
Scope: Fixed Q removes direct confounding by changing evaluation-domain difficulty. It contains repeated corruption versions of only 100 base images, however. There are no between-seed confidence intervals, and the 1,500 pairs are not 1,500 independent experimental replications.

4. C4-2: supervision, selection and full trajectories

Rates below aggregate raw numerators and denominators over the entire run, rather than weighting batches with different selection counts equally. Selected-class concentration is computed per block; its mean and range are reported separately.

Supervision diagnostic	$\lambda=0$	$\lambda=1$
Pseudo-label error: all presentations	46.85%	42.31%
Pseudo-label error: selected presentations	19.08%	12.84%
Selection coverage	43.09%	43.31%
Selected / total presentations	77,555 / 180,000	77,964 / 180,000
Teacher/student argmax disagreement	42.91%	39.98%
Mean selected-class concentration	0.0663	0.0439
Concentration range across blocks	0.0108–0.2929	0.0195–0.1429
Optimizer updates / empty batches	2,820 / 0	2,820 / 0

Observation: $\lambda=1$ has lower aggregate pseudo-label noise and similar coverage, yet higher final student Q-CE. A simple account based on higher average pseudo-label error therefore does not explain the endpoint difference. Error identity, confidence, timing and selection distributions may matter; their individual effects are not identified here.



C4-2. All blocks and both arms: performance, pseudo-label error, coverage, concentration and teacher-student disagreement.

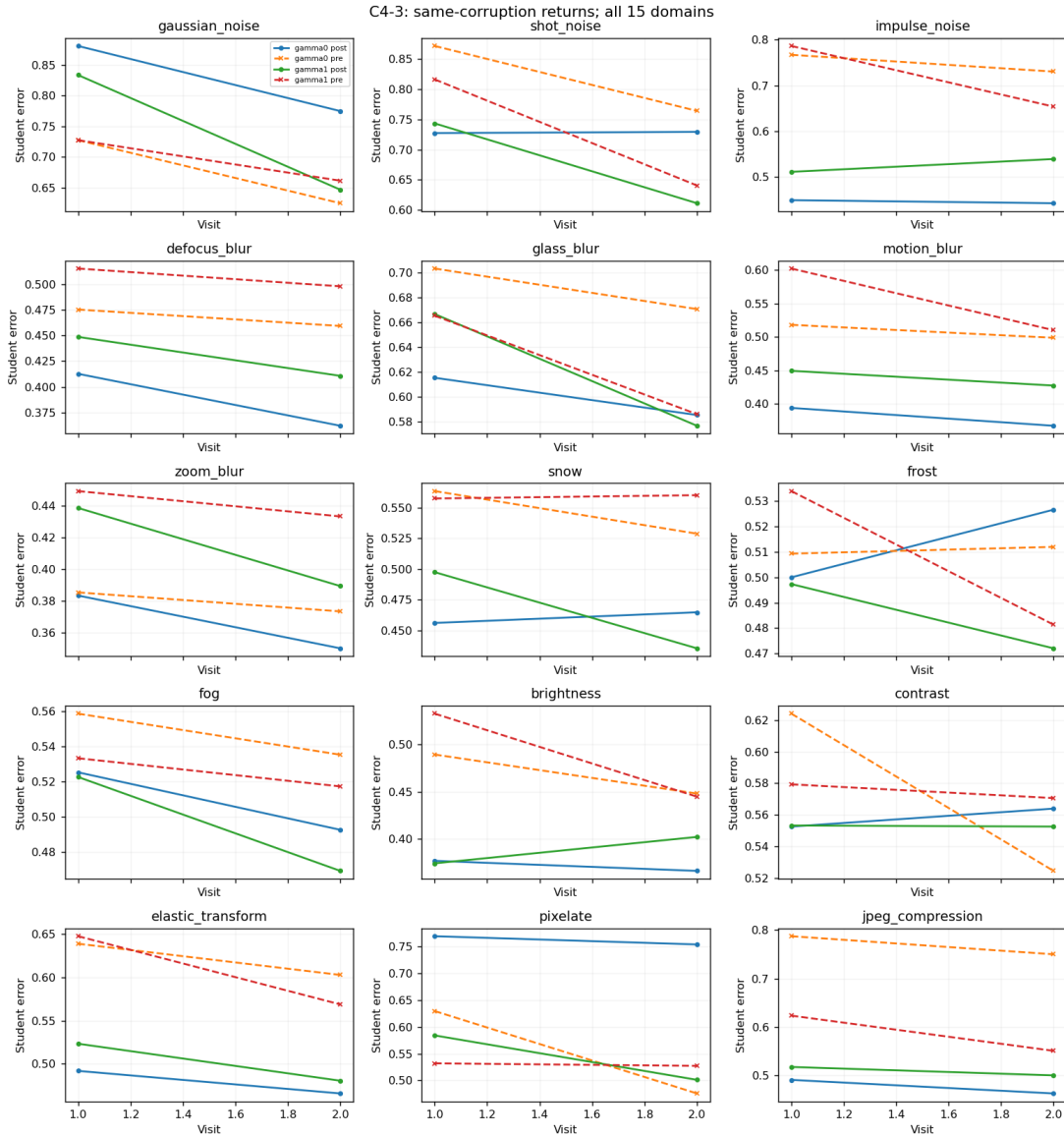
Current-domain risk varies partly because each block has a different corruption. Concentration measures label imbalance, not harmful collapse by itself. Teacher error is shown separately; teacher CE is retained in the block tables and reported on fixed Q in Section 3.

5. C4-3: returns to the same corruption

The return contrast is post-adaptation error on visit 2 minus visit 1, using the same corruption and audit IDs. Negative values indicate improvement; pp denotes percentage points.

At $\lambda=0$, the mean over 15 domains is -2.12 pp, with 11 improving and 4 worsening domains. At $\lambda=1$, it is -4.97 pp, with 13 improving and 2 worsening domains. These comparisons favor recovery/improvement overall. Two visits neither establish a long-term trend nor locate a turning point.

Adverse changes at $\lambda=0$ are frost $+2.67$, contrast $+1.13$, snow $+0.87$ and shot noise $+0.20$ pp. At $\lambda=1$, impulse noise and brightness each worsen by $+2.80$ pp. The largest improvements are on Gaussian noise: -10.60 and -18.67 pp, respectively. All 15 domains are shown below, including favorable outcomes.



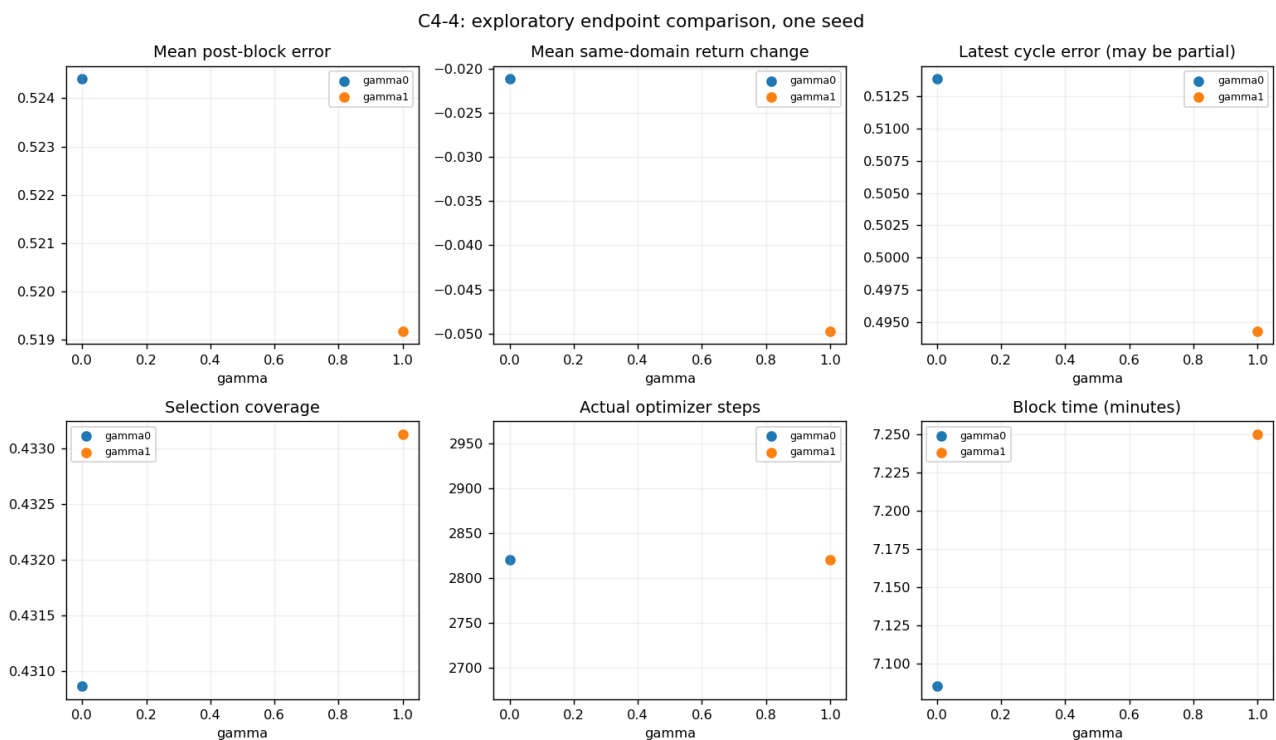
C4-3. All pre/post visit measurements for 15 corruptions. Dashed lines show entry performance; solid lines show post-adaptation performance.

6. C4-4: the parameter slice and actual cost

Only $\gamma=(0,0.001)$ and $(1,0.001)$ were run. Mean post-block risk averages performance immediately after adapting to each domain. It is neither the fixed-Q endpoint nor a reevaluation of the final model across all domains.

Measurement	$\lambda=0$	$\lambda=1$
Mean student post-error, all 30 blocks	52.44%	51.92%
Mean student post-error, cycle 1	53.50%	54.40%
Mean student post-error, cycle 2	51.38%	49.43%
Mean teacher post-error, all 30 blocks	41.13%	42.56%
Mean student post-CE, all 30 blocks	2.52751	2.58991
Mean teacher post-CE, all 30 blocks	1.70928	1.83773
Block computation time, two cycles	7.09 min	7.25 min

The corresponding mean current-domain source error is 45.68%, with CE 2.43707. Model choice, temporal aggregation and loss change the ranking. $\lambda=1$ is not uniformly worse than $\lambda=0$. Two λ endpoints do not identify a continuous feedback gain or a critical boundary.



C4-4. The two executed endpoints constitute a parameter slice, not a grid or phase diagram.

The natural pilot ran on two RTX 4090s in parallel. Block timings include audits/checkpoints, but exclude full setup and transport. Short replications and H1410 then ran on two Sapelo zhai_p L40S GPUs. The Slurm allocation lasted 25 min 35 s; each long fork took about 15.4–15.9 min, with parents running in parallel. L40S peak allocated/reserved memory was 2.093/2.800 GB per worker. Peak memory was not instrumented for the original 4090 pilot. Vast was destroyed; estimated total cost is \$1.0–1.2, not a final invoice.

7. Stage 2 / C4-5: isolating a feedback contrast

At checkpoints after blocks 5 and 15 of cycle 1, the first nonempty next-domain update was the candidate. Both nodes were specified before their outcomes and reported regardless of local sign. Accept/reject states include the complete student, teacher and optimizer; continuations share future image and augmentation keys.

A closed-loop continuation from the reject state first generates a reference tape: target probabilities, pseudo-labels, confidence masks and input keys. Accept/replay uses this external tape; its own teacher supplies neither training targets nor masks. Replay continues learning. It cuts the student's influence on future supervision; it does not freeze the learner or merely fix the already-exogenous image stream.

Let L denote student CE on fixed Q :

- $A_{cl}(h) = L_{accept,closed}(h) - L_{reject,closed}(h)$: the descendant effect of accepting under endogenous supervision.
- $A_{ol}(h) = L_{accept,replay}(h) - L_{reject,replay}(h)$: the effect under common external supervision.
- $I(h) = A_{cl}(h) - A_{ol}(h)$: the feedback interaction under this intervention. Positive values mean acceptance has a worse effect in closed loop.

Reject/replay and reject/closed were explicitly checked for equality over the first 20 steps. Their later equivalence follows from the deterministic recurrence, allowing reuse of the reject path. Three full branches were run; a fourth full branch was not independently executed.

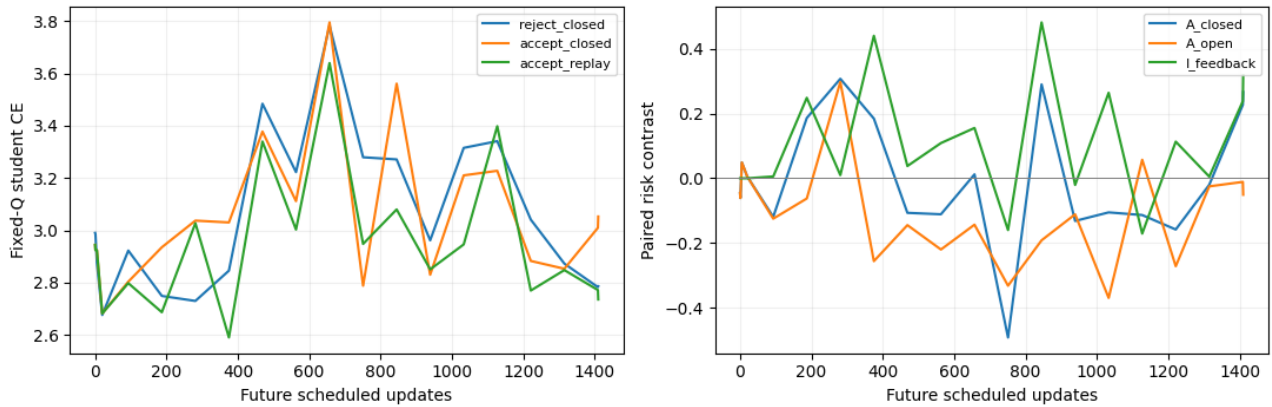
Parent	Future steps h	A_{cl}	A_{ol}	I
Block 5	0 (immediate)	-0.045932	-0.045932	0
Block 5	20	+0.005858	+0.005858	0
Block 5	94	-0.119639	-0.116541	-0.003098
Block 5	1,410	+0.266292	-0.050690	+0.316982
Block 15	0 (immediate)	+0.046558	+0.046558	0
Block 15	20	+0.036279	+0.036013	+0.000266
Block 15	94	+0.138922	+0.443354	-0.304432
Block 15	1,410	+0.069601	-0.052072	+0.121673

All table values are from Sapelo. H20/H94/H1410 share prefixes and are not independent replications. H94 contrasts agree between 4090 and L40S within 4.77×10^{-7} : numerical reproducibility does not increase the scientific sample size.

The block-5 short reversal is not attributable to feedback: at $h=5/20$, closed and replay acceptance effects coincide. At $h=94$, acceptance is beneficial again. The prespecified long endpoint exhibits the joint pattern of immediate benefit, adverse closed-loop descendants, and beneficial replay descendants.

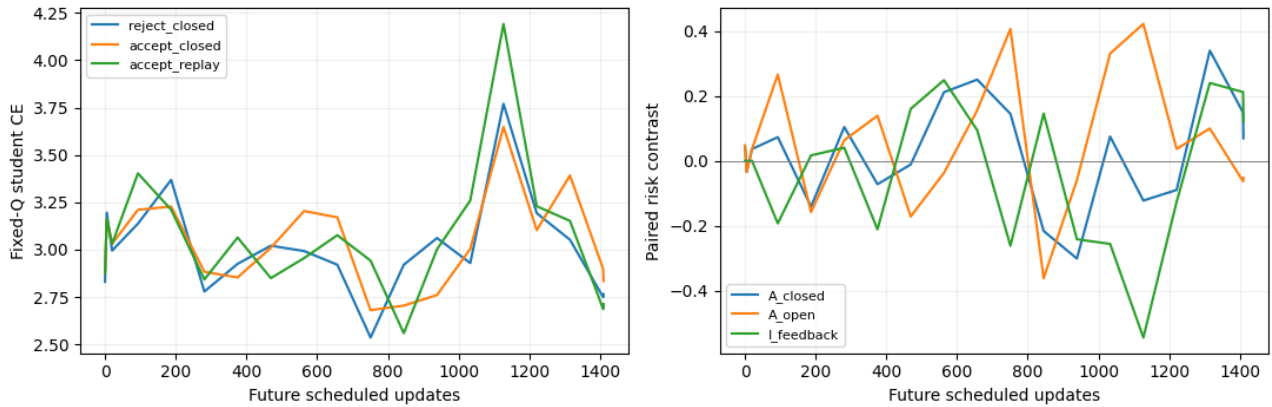
8. C4-5: full curves and interpretation

C4-5: prespecified fork after block 5; one parent, exploratory



Prespecified block-5 fork. Left: absolute fixed-Q CE. Right: acceptance effects and feedback interaction.

C4-5: prespecified fork after block 15; one parent, exploratory



Prespecified block-15 fork. All audit points are retained; the interaction repeatedly changes sign.

Both prespecified H1410 endpoints have positive I , but feedback is not uniformly harmful along either trajectory. Both H94 interactions are negative, with further sign changes at intermediate audits. The result concerns specific parents, supervision interventions and horizons; it is not a monotone law of error amplification.

H1410 equals $15 \text{ domains} \times 94 \text{ batches}$. Because the focal candidate consumes the first batch of its domain, the end of the fifteenth continuation domain block is at $h=1409$. Interactions there are also positive: $+0.237884$ and $+0.211858$. Both times are retained to distinguish a cycle-length budget from the exact domain-block boundary; neither removes the intermediate reversals.

Relative to the reject tape, accepted closed-loop continuations change $9,252 / 6,892$ pseudo-label events and $4,111 / 3,683$ mask events for blocks 5 / 15. These count repeated sample presentations. The intervention fixes targets and masks jointly, so it does not identify their separate contributions.

9. Absolute deterioration, relative harm and next steps

Absolute performance does not uniformly deteriorate

Parent / path	Initial CE	H1410 CE	Initial error	H1410 error
5 / reject closed	2.99013	2.78637	65.73%	55.27%
5 / accept closed	2.94420	3.05266	64.67%	56.73%
5 / accept replay	2.94420	2.73568	64.67%	51.33%
15 / reject closed	2.83018	2.76474	55.53%	50.47%
15 / accept closed	2.87674	2.83434	56.40%	52.13%
15 / accept replay	2.87674	2.71267	56.40%	53.07%

Every path improves error from its own initial state. Block 15 accept/closed even has lower endpoint error than accept/replay despite higher CE. Worse than rejecting, worse under feedback than replay, and continuously rising absolute risk are distinct claims. Selected prespecified comparisons support the first two, not the third.

Target claim	Current evidence
Proxy/local labelled disagreement	Observed on scheduled finite-sample probes
Immediate benefit, worse later descendants	Observed for block 5 on fixed-Q CE
Additional harm relative to specified replay	Observed at both H1410 endpoints; intermediate directions vary
Persistent recursive deterioration	Not established; returns generally improve and trajectories recover
Closed four-dimensional diagnostic dynamics	Untested; diagnostic measurements alone do not establish closure
Jacobian, spectral radius or phase boundary	Not estimated or validated; C4-6/7 remain unrun
Hypercompression or lossy inheritance	Not applicable to this implementation
General RSI result or benchmark superiority	Unsupported by one seed, one dataset and simplified CTTA

Recommended next experiment: replicate the frozen setup across seeds. Keep the two parent checkpoints, λ/β , H94/H1410 and both metrics; prespecify reporting of endpoints and complete audit trajectories. First assess reproducibility on the exploratory audit. If proceeding to Stage 4, freeze the primary contrast, meaningful effect threshold and analysis before using seeds 303/404/505/606/707 and the untouched confirmation audit. No success threshold has yet been established by these data.

If the endpoint interaction replicates, separate target and mask interventions. Stage 3 additionally requires independent student/teacher perturbations, scale and rank/conditioning checks, momentum-based closure tests and held-out response predictions. Fitting a VAR to the existing trajectories does not complete it. If replication is unstable, retain the negative findings and do not escalate to a persistent-deterioration claim.

10. Evidence, deliverables and provenance

This report presents verified outcomes aligned with integrated C4 v3. Evidence comes from `Delivery/RESULTS.md`, machine-readable summaries and fork completion/verification records. Preparing this English report involved no new GPU experiments.

Deliverable	Location / contents
Current protocol	<code>Protocol.tex/pdf</code> , integrated v3; hash matches <code>Downloads/C4.tex</code>
Pilot tables	<code>Delivery/results/report/blocks.json</code> , <code>updates.jsonl</code> , <code>local_sign_counts.json</code> , <code>summary.json</code>
C4-1 through C4-4	All five figures in Sections 2–6, including fixed-Q figure C4-2b
C4-5	<code>Delivery/sapelo_results/after{5,15}_h{20,94,1410}/</code> : comparisons, metrics, runtime and verification
Unrun deliverables	C4-gated, C4-6 dynamics, C4-7 parameter map and independent confirmation
Full intermediate evidence	Batch predictions, pseudo-labels/probabilities, masks, IDs, augmentation keys, complete checkpoints, failed attempts and logs
Repository	<code>mivolis/RSI2RSD</code> , branch <code>experiment/C4</code> , directory <code>C4/</code> ; main unchanged

Sapelo job 48416583 completed with exit 0:0. Each H1410 fork passed verification of 4,364 hashed files and 4,230 future batch records. Canonical outputs are at `/scratch/yz54720/RSI2RSD/C4/migration_results/job48416583/`. All 7,250 required Vast files were recovered and verified before instance destruction.

Computation and artifact publication have separate statuses. Summaries, figures, configurations and verification records were delivered in result commit `790d6a68bca12ee0ffbc98196a31b63a91b6c677`. Bulk attachments were still in the upload/mirroring workflow at report preparation; a compact-results commit does not mean all raw files are downloadable. Current availability is tracked in `ARTIFACTS.md`.

Repository entry point:

<https://github.com/mivolis/RSI2RSD/tree/experiment/C4/C4>

Suggested meeting conclusion

In this natural CTTA learner, decreasing the pseudo-label objective does not guarantee improved local labelled risk. A prespecified fork also provides an example of an immediately beneficial update with worse closed-loop descendants at a specified long horizon; common-supervision replay links that descendant contrast to the endogenous supervision channel. However, effects change sign over time, CE and error disagree, and same-domain returns improve overall. The next justified step is to replicate and delimit this finite-horizon phenomenon, rather than claim persistent self-deterioration or a phase transition.